

DIRECTORATE GENERAL OF MINES SAFETY (DGMS)

Junior Scientific Officer – GEOLOGY

Examination 2019 | 100 Questions

Note: Each question carries equal marks. Choose the most appropriate answer from the options given.

SECTION A: SEDIMENTOLOGY, PETROLOGY & METAMORPHISM

Q.1. A normal graded bed of sandstone is noticed with flute cast at its base and internally made up of massive, plane-laminated and ripple-laminated subdivisions, in order of superposition. The sandstone bed is interpreted as a product of

- (a) Debris flow
- (b) Grain flow
- (c) Turbidity current
- (d) Contour current

Q.2. In carbonate diagenesis, if calcite cement is overgrown on a single calcite crystal in same optical orientation, it is referred to as

- (a) Rim overgrowth
- (b) Antitaxial overgrowth
- (c) Blocky overgrowth
- (d) Syntaxial overgrowth

Q.3. A carbonate rock that contains almost no mud and shows grain-supported framework is named as

- (a) Grainstone
- (b) Packstone
- (c) Floatstone
- (d) Bindstone

Q.4. In a Barrovian zone of metamorphism, presence of biotite + chlorite + phengitic muscovite + quartz + albite assemblage marks the

- (a) Albite zone
- (b) Biotite zone
- (c) Zeolite zone
- (d) Chlorite zone

Q.5. Which one of the following is a characteristic assemblage of granulite facies metamorphism of mafic protolith?

- (a) Hornblende + Plagioclase + Quartz + Garnet
- (b) Forsterite + Orthopyroxene + Clinopyroxene + Epidote
- (c) Clinopyroxene + Orthopyroxene + Plagioclase + Garnet + Quartz
- (d) Forsterite + Orthopyroxene + Clinopyroxene + Staurolite

Q.6. When flattening occurs in rocks that contain rigid grains, zones of relatively low pressure develop on the sides of the crystals that are normal to the maximum compressive stress. These zones are called

- (a) Solution cleavage
- (b) Pressure shadows
- (c) Pressure solution
- (d) Stylolites

Q.7. Match List-I (Rock) with List-II (Description): A. Blueschist — B. Pseudotachylite — C. Hornfels — D. Skarn
1. Ultrafine-grained vitreous looking material occurring as thin veins
2. Na-amphibole containing schistose rock
3. A non-schistose substantial recrystallization
4. A metasomatic Ca-Fe-Mg-Mn-silicate rock with compositional zones and bands

- (a) A-4, B-3, C-1, D-2

- (b) A-2, B-1, C-3, D-4
- (c) A-2, B-3, C-1, D-4
- (d) A-3, B-4, C-1, D-2

Q.8. Presence of calcite, dolomite, quartz, diopside, wollastonite, K-feldspar, forsterite in a regionally metamorphosed rock, under granulite facies suggests which one of the following types of protoliths?

- (a) Mafic
- (b) Pelitic
- (c) Carbonate rock
- (d) Ultramafic

Q.9. $3 \text{ Anorthite} = \text{Grossular} + 2 \text{ Kyanite} + \text{Quartz}$ is a reaction which marks the eclogite facies in metabasic rocks. What will be the composition of grossular in the reaction given above?

- (a) $\text{Ca}_3\text{Al}_2\text{Si}_3\text{O}_{12}$
- (b) CaSiO_3
- (c) $(\text{Ca, Mg})\text{Si}_2\text{O}_6$
- (d) CaCO_3

Q.10. Granoblastic polygonal texture is an example of

- (a) Texture of recrystallization
- (b) Texture of crystallization
- (c) Texture of metasomatism
- (d) Texture of mylonitization

Q.11. In metamorphism of calc-silicate rocks, which assemblage can be considered equivalent to the sillimanite zone of pelitic rocks?

- (a) Anorthite-pyroxene
- (b) Zoisite-calcite-biotite
- (c) Diopside-anorthite
- (d) Wollastonite-grossular

SECTION B: STRUCTURAL GEOLOGY & TECTONICS

Q.12. The hinge line of a fold is

- (a) A line that divides the fold into symmetrical parts
- (b) A line that is always parallel to fold axis
- (c) A line of maximum curvature in a folded bed
- (d) A line of minimum curvature in a folded bed

Q.13. In ocean, regional strike-slip faults associated with lithospheric plate boundaries are known as

- (a) Transform faults
- (b) Tear faults
- (c) Wrench faults
- (d) Scissor faults

Q.14. In a fold, rock salt has broken through the more brittle overlying rocks and occupied its core. The structure is known as

- (a) Disharmonic fold
- (b) Drag fold
- (c) Box fold
- (d) Diapir fold

Q.15. A fault having both dip-slip as well as strike-slip components is termed as

- (a) Strike-slip fault
 - (b) Dip-slip fault
 - (c) Oblique-slip fault
 - (d) Wrench fault
-

SECTION C: MINERALOGY & CRYSTALLOGRAPHY

Q.16. Four three-fold axes, each inclined at $54^{\circ}44'$ to the crystallographic axes, are characteristic symmetry of which one of the following crystal systems?

- (a) Hexagonal
- (b) Isometric
- (c) Orthorhombic
- (d) Triclinic

Q.17. Ca-rich plagioclase breaks down to almost pure albite and epidote during which one of the following deuteric alterations?

- (a) Saussuritization
- (b) Devitrification
- (c) Chloritization
- (d) Uralitization

Q.18. Which one of the following rocks is normally expected to have the highest abundance of incompatible trace elements?

- (a) Basalt
- (b) Dunite
- (c) Peridotite
- (d) Granite

Q.19. Which one of the following magma series is found only in subduction zones?

- (a) Calc-alkaline
- (b) Tholeiitic
- (c) Alkaline
- (d) Komatiite

SECTION D: GEOCHEMISTRY & GEOCHRONOLOGY

Q.20. Average continental crust reservoir is represented by which one of the following Rb/Sr values of radiogenic parent-daughter pairs?

- (a) 0.124
- (b) 0.158
- (c) 0.182
- (d) 0.242

Q.21. In a rare earth element (REE) diagram, europium anomaly is seen between which one of the following REE pairs?

- (a) La-Nd
- (b) Nd-Sm
- (c) Sm-Gd
- (d) Gd-Dy

Q.22. Which one of the following symbols denotes the deviation of the Nd-isotopic composition of a rock from that of the chondritic uniform reservoir?

- (a) ω (Omega)
- (b) ϵ (Epsilon)
- (c) δ (Delta)
- (d) υ (Upsilon)

Q.23. In a $^{238}\text{U}/^{206}\text{Pb}$ versus $^{235}\text{U}/^{207}\text{Pb}$ ratio diagram, the locus of points having the same $^{238}\text{U}/^{206}\text{Pb}$ and $^{235}\text{U}/^{207}\text{Pb}$ ages is known as

- (a) Mixing line
- (b) Isochron line
- (c) Concordia curve

(d) Discordia curve

Q.24. In the atomic structure of silicate melts, Ca^{2+} and Mg^{2+} ions are categorized as

- (a) Network modifiers
- (b) Network formers
- (c) Bridging atoms
- (d) Non-bridging atoms

Q.25. Which one of the following Mid-Oceanic Ridge Basalts (MORB) is derived from an incompatible element-poor mantle?

- (a) N-MORB
- (b) P-MORB
- (c) T-MORB
- (d) VAB

Q.26. Which one of the following geochemical reservoirs is defined by the Nd-isotopic composition of chondritic meteorites?

- (a) DM
- (b) BSE
- (c) PREMA
- (d) HIMU

Q.27. Which of the following do NOT constitute an equilibrium mineral assemblage in a rock?

- (a) Quartz and orthoclase
- (b) Forsterite and quartz
- (c) Hornblende and andesine
- (d) Labradorite and augite

SECTION E: SEDIMENTARY ENVIRONMENTS & STRATIGRAPHY

Q.28. In a wave-dominated coastline, the correct distribution of environment from land to sea is

- (a) Shoaling zone–surf zone–breaker zone–swash zone
- (b) Swash zone–shoaling zone–surf zone–breaker zone
- (c) Swash zone–surf zone–breaker zone–shoaling zone
- (d) Surf zone–shoaling zone–swash zone–breaker zone

Q.29. The uppermost part of subaqueous delta that lies at water depth down to 10 m or so, is referred to as

- (a) Upper delta plain
- (b) Lower delta plain
- (c) Prodelta
- (d) Delta front

Q.30. Which one of the following sequences of sedimentary structures is correct in terms of increasing bed shear?

- (a) Parting lineation, dune, flute mark, ripple
- (b) Ripple, dune, parting lineation, flute mark
- (c) Dune, parting lineation, flute mark, ripple
- (d) Flute mark, parting lineation, ripple, dune

Q.31. The deep-focused earthquakes commonly originate along

- (a) Mid-oceanic ridge
- (b) Continental rift
- (c) Transform fault
- (d) Benioff zone

Q.32. A major continent that existed before 90 Ma and comprised present-day N. America, Greenland, Europe and part of Asia is referred to as

- (a) Superia
- (b) Gondwanaland
- (c) Laurentia
- (d) Laurasia

Q.33. The rocks and sediments that were subducting sometimes get attached with overriding plates by thrusting. This process is commonly called

- (a) Obduction
- (b) Rejuvenation
- (c) Underplating
- (d) Offscraping

Q.34. The lithosphere is comprised of

- (a) Mantle and outer core
- (b) Mantle and asthenosphere
- (c) Upper mantle and crust
- (d) Continental and oceanic crusts

Q.35. In hexagonal crystals, there is one and only one direction which shows no double refraction of light. This direction is called

- (a) Optic axis
- (b) Optic axial plane
- (c) Isogyre
- (d) Optic normal

Q.36. The interference figure produced by a uniaxial crystal cut parallel to its optic axis is known as

- (a) Dichroic figure
- (b) Flash figure
- (c) Brewster's figure
- (d) Uniaxial indicatrix

Q.37. Which one of the following instruments can be used to distinguish kyanite, sillimanite and andalusite?

- (a) EPMA
- (b) XRD
- (c) XRF
- (d) ICP-MS

Q.38. Match List-I (Mineral example) with List-II (Silicate class): A. Phlogopite — B. Enstatite — C. Anthophyllite — D. Beryl 1. Inosilicate single chain 2. Cyclosilicate 3. Phyllosilicate 4. Inosilicate double chain

- (a) A-3, B-1, C-4, D-2
- (b) A-2, B-1, C-4, D-3
- (c) A-2, B-4, C-1, D-3
- (d) A-3, B-4, C-1, D-2

Q.39. Muscovite belongs to which one of the following types of phyllosilicate structures? (o = octahedral, t = tetrahedral)

- (a) Dioctahedral t-o-t, interlayer cation, dioctahedral t-o-t type
- (b) Dioctahedral t-o-t type only
- (c) Trioctahedral t-o-t, interlayer cation, trioctahedral t-o-t type
- (d) Dioctahedral t-o type only

Q.40. Which one of the following expressions is correct as per notations of Miller indices of a crystal face?

- (a) (1 1 ■)
- (b) ($\frac{1}{2}$ $\frac{1}{2}$ ■)
- (c) (1a 1b 1c)

(d) 1 1 1

Q.41. The exsolution texture in which the host mineral is a plagioclase feldspar and the lamellae are of KAlSi_3O_8 composition is known as

- (a) Perthite
- (b) Antiperthite
- (c) Myrmekite
- (d) Inverted pigeonite

Q.42. Match List-I (Mineral name) with List-II (Mineral classification): A. Sylvite — B. Manganite — C. Siderite — D. Monazite 1. Carbonate 2. Phosphate 3. Hydroxide 4. Halide

- (a) A-4, B-3, C-1, D-2
- (b) A-1, B-3, C-4, D-2
- (c) A-4, B-2, C-3, D-1
- (d) A-3, B-2, C-4, D-1

Q.43. 'Labradorescence' observed in plagioclase is caused by

- (a) Polymorphism
- (b) The presence of closely spaced lamellae of An_{10} and An_{90} composition
- (c) Fine exsolution lamellae of An_{10} to An_{90} composition
- (d) Peristerite gap

Q.44. Match Hermann-Mauguin notations List-I with List-II (Crystal class): A. $3A_2$, $4A_2$, $6A_2$ — B. $1A_1$, $4A_1$ — C. $1A_1$, $2m$ — D. $1A_1$, $6A_2$ 1. $4\ 2\ 2\ 2$ 2. $4\ 3\ 2\ 3$ 3. $m\ m\ 2\ 4$ 4. $6\ 2\ 2$

- (a) A-2, B-1, C-3, D-4
- (b) A-1, B-2, C-4, D-3
- (c) A-4, B-3, C-2, D-1
- (d) A-3, B-4, C-1, D-2

Q.45. In each crystal class there is a form, the faces of which intersect all the crystallographic axes at different lengths. This form is known as

- (a) Special form
- (b) General form
- (c) Open form
- (d) Closed form

Q.46. Which one of the following pairs regarding weathering is NOT correctly matched?

- (a) Hydration : Occurs when minerals absorb water molecules
- (b) Oxidation : Occurs when an atom or an ion gains an electron
- (c) Carbonation : Occurs when carbonates are formed
- (d) Chelation : Occurs when metal ion is removed by organic matter

Q.47. The mean age of the oceanic floor is approximately

- (a) 20 Ma
- (b) 60 Ma
- (c) 80 Ma
- (d) 230 Ma

Q.48. According to Airy's hypothesis, if the thickness of crustal column is doubled, the mountain heights may increase approximately by

- (a) 80%
- (b) 40%
- (c) 30%
- (d) 20%

Q.49. Match List-I (Volcano type) with List-II (Characteristic): A. Shield volcano — B. Cinder cone volcano — C. Stratovolcano — D. Lava dome volcano 1. Also known as composite volcano 2. Has shape of a flattened dome 3. Formed exclusively of pyroclastic materials 4. Formed by highly viscous lava

- (a) A-4, B-1, C-3, D-2

- (b) A-4, B-3, C-1, D-2
- (c) A-2, B-3, C-1, D-4
- (d) A-2, B-1, C-3, D-4

Q.50. Match List-I (Bed form in sandy alluvial channel) with List-II (Froude number, F): A. Dunes with superimposed ripples — B. Washed out dunes — C. Plane bed — D. Ripples 1. $F \ll 1$ 2. $F \geq 1$ 3. $F \approx 1$ 4. $F = 1$

- (a) A-1, B-2, C-4, D-3
- (b) A-1, B-4, C-2, D-3
- (c) A-3, B-2, C-4, D-1
- (d) A-3, B-4, C-2, D-1

SECTION F: GEOMORPHOLOGY, HYDROGEOLOGY & ENGINEERING GEOLOGY

Q.51. Which one of the following pairs is NOT correctly matched?

- (a) Nunatak : Unglaciaded island of bedrock, formerly surrounded by ice
- (b) Cirque : Steep-walled semicircular basin in mountain glaciers
- (c) Col : Narrow, sharp-edged ridge in mountain glaciers
- (d) Fjord : U-shaped glacial trough below sea level

Q.52. The lateral unsaturated flow of water in the soil zone, where a highly permeable geological unit overlays a less permeable geological unit, is termed as

- (a) Overland flow
- (b) Saturated overland flow
- (c) Sheet flow
- (d) Throughflow

Q.53. The void ratio of the solid mass is defined as

- (a) Volume of voids / Total volume
- (b) Volume of voids / Volume of solids
- (c) Volume of solids / Volume of voids
- (d) Volume of solids / Total volume

Q.54. Which one of the following pairs is NOT correctly matched?

- (a) Annular drainages : Develop on eroded dome in alternate resistant/weak sediments
- (b) Parallel drainages : Develop on horizontal sediments or homogeneous crystalline rocks
- (c) Trellis drainages : Develop on alternating weak and resistant tilted sedimentary rocks
- (d) Discordant drainages : Develop on river channels cut across geological structures

Q.55. Along which of the following plate boundaries, the moving plates grind past each other without producing or destroying lithosphere?

- (a) Diverging plate boundary
- (b) Converging plate boundary only
- (c) Transform fault plate boundary only
- (d) Converging plate boundary and transform fault plate boundary both

Q.56. Stress that is generated due to the weight of the overlying rock column in the earth's crust is referred to as

- (a) Confining pressure
- (b) Tensile pressure
- (c) Lithostatic pressure
- (d) Deviatoric pressure

Q.57. The separation of a competent layer due to layer-parallel extension is incomplete and shows a narrowing of the segments. This structure is termed as

- (a) Pinch-and-swell structure
- (b) Chocolate-tablet structure

- (c) Extension fracture boudinage
- (d) Tension gash

Q.58. Overfolds have one of the limbs essentially

- (a) Horizontal
- (b) Inverted
- (c) Vertical
- (d) Parallel to the other

Q.59. The ocean that gave rise to birth of the Himalayan mountains was

- (a) Indian Ocean
- (b) Pacific Ocean
- (c) Tethys Ocean
- (d) Panthalassa Ocean

Q.60. A continental margin that forms due to initial rifting of a continent to develop an ocean later, is called

- (a) Passive margin
- (b) Active margin
- (c) Sutured margin
- (d) Andean-style margin

SECTION G: SEISMOLOGY & EARTH INTERIOR

Q.61. Palaeoseismicity is related to the study of

- (a) Earthquakes of the geological past through slippage observed in the field along faults
- (b) Relationship between earthquake and faulting
- (c) Mapping of earthquakes
- (d) Risk related to seismic activities

Q.62. Consider the following statements regarding the earth's interior: 1. Seismic waves reveal that the average depth of mantle is about 3100 km. 2. Based on the seismic wave studies, the concept of solid inner core was given by Inge Lehmann. 3. The sharp contrast from seismological wave velocities below the crust was noticed by Andrija Mohorovičić. 4. Transverse waves do not propagate in the outer core and hence the material here is in fluid stage. Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2, 3 and 4 only
- (c) 1, 3 and 4 only
- (d) 1, 2, 3 and 4

Q.63. Consider the following statements regarding Richter scale: 1. It is the first scale to estimate the size of an earthquake by instrument. 2. It is based on the released energy at focus of the earthquake. 3. It is based on the maximum amplitude of waves measured on a seismograph. 4. It is based on extent of the damage done by the seismic waves. Which of the statements given above is/are correct?

- (a) 1 and 3
- (b) 2 and 3
- (c) 1 and 4
- (d) 2 and 4

Q.64. The velocity of the seismic waves within the earth ranges from

- (a) 1 to 4 km/second
- (b) 2 to 8 km/second
- (c) 3 to 10 km/second
- (d) 2 to 13 km/second

Q.65. Which one of the following is the correct order of abundance of dissolved salts in ocean water?

- (a) NaCl > MgSO₄ > MgCl₂ > CaCO₃ > K₂SO₄ > CaSO₄ > MgBr₂

- (b) $\text{NaCl} > \text{MgCl}_2 > \text{MgSO}_4 > \text{CaSO}_4 > \text{K}_2\text{SO}_4 > \text{CaCO}_3 > \text{MgBr}_2$
 (c) $\text{NaCl} > \text{K}_2\text{SO}_4 > \text{MgSO}_4 > \text{MgCl}_2 > \text{CaCO}_3 > \text{CaSO}_4 > \text{MgBr}_2$
 (d) $\text{NaCl} > \text{MgCl}_2 > \text{K}_2\text{SO}_4 > \text{CaSO}_4 > \text{CaCO}_3 > \text{MgBr}_2 > \text{MgSO}_4$

SECTION H: ICHNOLOGY, REMOTE SENSING & GIS

Q.66. Zoophycos assemblage is the characteristic ichnofacies of which one of the following zones?

- (a) Sublittoral zone
- (b) Bathyal zone
- (c) Abyssal zone
- (d) Littoral zone

Q.67. Which one of the following errors results in because of wind effects in aerial photography?

- (a) Drift
- (b) Crab
- (c) Tilt
- (d) Relief displacement

Q.68. In a GIS database, the location of meteorological station with respect to a specific land use can be identified using

- (a) Line to polygon overlay
- (b) Polygon on polygon overlay
- (c) Point in polygon overlay
- (d) Line on line overlay

Q.69. GPS satellites carry rubidium and cesium atomic clocks that usually have time accuracy of 1 part in

- (a) 10^2 and 10^3 respectively
- (b) 10^2 and 10^8 respectively
- (c) 10^6 and 10^{10} respectively
- (d) 10^{12} and 10^{13} respectively

SECTION I: HYDROGEOLOGY & ENVIRONMENTAL GEOLOGY

Q.70. Consider the following statements: Statement 1: Passive microwave remote sensing can help in detecting seepage zones, springs, faults, soil moisture variations, etc. Statement 2: Water and vegetation have dielectric constants that are very different from those of other ground objects. Which one of the following is correct in respect of the above statements?

- (a) Both Statement 1 and Statement 2
- (b) Both Statement 1 and Statement 2 are true and Statement 2 is not the correct explanation of Statement 1
- (c) Statement 1 is true but Statement 2 is false
- (d) Statement 1 is false but Statement 2 is true

Q.71. Consider the following statements regarding specific yield of an aquifer: 1. It is the ratio of the volume of water that after saturation can be drained by gravity to its own volume. 2. Its value does not depend on the grain size. Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Q.72. The intrinsic permeability of the porous medium depends on the

- (a) Size of the opening through which the fluid moves only
- (b) Shape of the opening through which the fluid moves only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

Q.73. Which of the following statements regarding the Total Dissolved Solids (TDS) and Electrical Conductance (EC) of the groundwater are correct? 1. A rapid determination of the TDS can be made by measuring the EC of the groundwater sample. 2. The specific conductance is measured in microsiemens/cm. 3. The specific conductance decreases with increase in temperature.

- (a) 2 and 3 only
- (b) 1 and 3 only
- (c) 1 and 2 only
- (d) 1, 2 and 3

Q.74. Consider the following statements regarding irrigation water: Statement 1: Sodium concentration is important for classifying irrigation water. Statement 2: Sodium reacts with soil to reduce its permeability and hence affects the agricultural activity. Which one of the following is correct in respect of the above statements?

- (a) Both Statement 1 and Statement 2
- (b) Both are true but Statement 2 is not the correct explanation of Statement 1
- (c) Statement 1 is true but Statement 2 is false
- (d) Statement 1 is false but Statement 2 is true

Q.75. Consider the following statements regarding an aquifer composed of unconsolidated materials: 1. The resistivity increases with the degree of saturation. 2. The resistivity increases with increase in salinity of groundwater. Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) Both 1 and 2
- (d) Neither 1 nor 2

SECTION J: ORE GEOLOGY & ECONOMIC GEOLOGY

Q.76. Which one of the following statements regarding supergene copper deposit is NOT correct?

- (a) The top zone consisting of goethitic/limonitic mass with some clay is gossan.
- (b) The zone of secondary enrichment lies below the groundwater surface where copper content increases.
- (c) The supergene enrichment zone gradually merges with the protore.
- (d) The secondary enrichment zone contains native copper and chalcocite.

Q.77. Consider the following statements: 1. Algoma-type iron formations are sedimentary units in sequences of submarine volcanic rocks in greenstone belts, most commonly of Cambrian age. 2. Superior-type iron formations occur in sedimentary basins of marine strata of late Archaean or Palaeoproterozoic age. Which of the statements given above is/are correct?

- (a) 1 only
- (b) Both 1 and 2
- (c) 2 only
- (d) Neither 1 nor 2

Q.78. Match List-I (Hydrocarbon deposit) with List-II (Location): A. Lignite — B. Oil — C. Bituminous coal — D. Natural gas 1. Kalol, Gujarat 2. Neyveli, Tamil Nadu 3. Jaisalmer, Rajasthan 4. Kamte, Maharashtra

- (a) A-2, B-1, C-4, D-3
- (b) A-1, B-2, C-3, D-4
- (c) A-2, B-4, C-3, D-1
- (d) A-3, B-1, C-4, D-2

SECTION K: ENGINEERING & APPLIED GEOLOGY

Q.79. Which one of the following is NOT considered for the calculation of rock mass rating?

- (a) Uniaxial compressive strength
- (b) Condition of discontinuities

- (c) Rock quality designation
- (d) Slake durability index

Q.80. The suitable geological conditions required for construction of a dam are resistant and stable bedrock: 1. Foundation rock with downstream dip 2. Foundation rock with gentle upstream dip 3. Foundation rock with gentle upstream dip 4. Broad valley with faulted strata

- (a) 1 and 2
- (b) 3 and 4
- (c) 1 and 3
- (d) 1 and 4

Q.81. Which one of the following pairs is NOT correctly matched?

- (a) Nunatak : Unglaciated island of bedrock, formerly surrounded by ice
- (b) Cirque : Steep-walled semicircular basin in mountain glaciers
- (c) Col : Narrow, sharp-edged ridge in mountain glaciers
- (d) Fjord : U-shaped glacial trough below sea level

Q.82. The Atterberg limits that define the boundary between the plastic and semi-solid states is

- (a) Liquid limit
- (b) Plastic limit
- (c) Shrinkage limit
- (d) Zero% water content and plastic limit

Q.83. In which one of the following dams, the major portion of the water pressure is transmitted to the abutments or sides of the valley?

- (a) Earth dam
- (b) Arch dam
- (c) Buttress dam
- (d) Rock-fill dam

Q.84. The Brazilian test is used to determine

- (a) Crushing strength
- (b) Tensile strength
- (c) Transverse strength
- (d) Shear strength

Q.85. Which one of the following is the best condition for tunnel site consideration?

- (a) Tunnel along the axis of an anticline
- (b) Tunnel within thick horizontal formation
- (c) Tunnel along the axis of a syncline
- (d) Horizontal tunnel parallel to the dip direction of rock strata

SECTION L: ENVIRONMENTAL & MARINE GEOLOGY

Q.86. Consider the following statements regarding geothermal energy: 1. Most areas where this energy is being tapped are along or near the plate boundaries. 2. Seasonal variation in the earth's surface temperature significantly affects this energy. 3. The places on the earth where groundwater circulates near magma are the easier sites of extraction of this energy. 4. The places on the earth having geothermal gradient around 25–50°C/km are considered as good sites. Which of the statements given above is/are correct?

- (a) 1 and 3 only
- (b) 2 and 4 only
- (c) 1, 3 and 4 only
- (d) 1, 2, 3 and 4

Q.87. Carbon dioxide emissions from land entering the oceans: 1. Make oceanic water acidic 2. Make oceanic water basic 3. Help some marine organisms to make their carbonate shells by drawing

carbonate from the seawater

- (a) 1 only
- (b) 2 and 3
- (c) 2 only
- (d) 1 and 3

Q.88. In coastal areas, groundwater resource is being threatened by the encroachment of saltwater because: 1. Freshwater being less dense than saltwater, floats over the latter and forms large lens-shaped body 2. Saltwater being dense, is available at very shallow depths in coastal regions 3. Heavy pumping in coastal areas causes encroachment of saltwater 4. Waves and tides cause mixing of saltwater with freshwater

- (a) 1 and 3 only
- (b) 2 and 4 only
- (c) 1, 3 and 4 only
- (d) 1, 2, 3 and 4

Q.89. The organic matter load in a water body is described as

- (a) Dissolved oxygen
- (b) Biochemical oxygen demand
- (c) Chemical oxygen demand
- (d) Total hardness

SECTION M: GEOPHYSICS & REMOTE SENSING

Q.90. Which of the following are the requirements for locating epicentre of an earthquake? 1. Time difference data of first arrivals of P and S waves on a seismograph station 2. Highest amplitude data from seismograph stations 3. Time and amplitude data from several seismograph stations

- (a) 1, 2 and 3
- (b) 1 and 3 only
- (c) 2 and 3 only
- (d) 1 and 2 only

Q.91. The amount of radiant energy at wavelengths (λ) incident upon a surface per unit area of that surface is known as

- (a) Radiant energy (Q_λ)
- (b) Radiant flux (Φ_λ)
- (c) Irradiance (E_λ)
- (d) Radiant exitance (M_λ)

Q.92. Consider the following statements regarding reflected light: 1. The reflected light, vibrating in only one plane, is called polarized light. 2. The light reflected from a water body into the field of view of a camera is polarized light. 3. The light from a blue sky is non-polarized because it is reflected from non-metallic particles in the atmosphere. Which of the statements given above is/are correct?

- (a) 1 and 2
- (b) 2 and 3
- (c) 3 only
- (d) 1 only

Q.93. In the satellite images, when other factors are same

- (a) Equal curvature of the earth is manifested by all altitude sensors
- (b) Least curvature of the earth is manifested by highest altitude sensor
- (c) Least curvature of the earth is manifested by lowest altitude sensor
- (d) Highest curvature of the earth is manifested by lowest altitude sensor

Q.94. Nuclides heavier than ^{209}Bi are produced by which of the following nucleosynthetic processes?

- (a) Slow neutron capture (s-process) and p-p chain
- (b) Rapid neutron capture (r-process) and CNO cycle

- (c) Rapid neutron capture (r-process) and rapid proton capture (p-process)
- (d) Spallation of C and O nuclides (x-process)

Q.95. The chemical formula of which one of the following minerals can be expressed as a combination of nepheline and quartz?

- (a) Anorthoclase
- (b) Analcime
- (c) Antigorite
- (d) Albite

Q.96. Halmyrolysis refers to a process where

- (a) Addition of La takes place in magma during crustal contamination
- (b) Removal of Mg from magma takes place during crystal fractionation
- (c) Removal of some ions of river water takes place when the river enters the sea
- (d) Removal of Eu takes place during magmatic crystallization

Q.97. The initial $^{87}\text{Sr}/^{86}\text{Sr}$ ratio of the earth has been estimated from which one of the following?

- (a) Granite
- (b) Kimberlite
- (c) Chondrite
- (d) Achondrite

Q.98. Which one of the following is a chalcophile element?

- (a) Oxygen
- (b) Lithium
- (c) Silver
- (d) Niobium

Q.99. Which one of the following statements regarding geophysical exploration methods is NOT correct?

- (a) Milligal is a unit of magnetic field.
- (b) Magnetic anomaly can be caused by pyrrhotite and magnetite.
- (c) Gravity method of mineral exploration is based on density contrasts.
- (d) Scintillometers can detect radioactivity in rocks.

Q.100. Which one of the following can be termed as significant geochemical anomaly in mineral exploration?

- (a) Concentration of indicator elements lower than the local threshold
- (b) Concentration of indicator elements falling within the regional threshold
- (c) Concentration of indicator elements falling within the local threshold
- (d) Concentration of indicator elements higher than the local background